

Jack Pellen-Pickersgill

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I'm a developer and teacher with a proven understanding of machine learning and AI. I thrive in challenging, collaborative environments, especially those which provide opportunities for exploring new tools and technologies. I have excellent research skills having worked on several highly technical projects as well as expert Python proficiency and experience working in an agile setting.

Languages

Python SQL Elixir Elm JavaScript HTML/CSS Prolog Java C# Bash R Matlab
Rust

AI and Machine Learning

Reinforcement Learning SAT/CSP AI Planning Evolutionary Game Theory Q-Learning
Agent Based Modelling Natural Language Processing Time Series Analysis Deep Learning
Clustering

Tools and Libraries

LaTeX Numpy Postgres SvelteJS Matplotlib Regex Linux PyTest Z3 OTP
Vim/VimScript Flask SQLite Pandas Hypothesis Cargo

Methodologies

Scrum/Agile Distributed Systems Version Control Unit Testing RESTful APIs Data Viz.
Documentation

Work Experience

- **STEM Foundation Programming Tutor** - Royal Holloway University, 2022-present
 - Took on a more leading role running Python programming labs for first year STEM students of mixed backgrounds.
 - Responsibilities also include monitoring student wellbeing, holding pastoral seminars and offering one-to-one sessions for students who need extra help.
- **Computer Science Teaching Assistant** - Royal Holloway University, 2021-present
 - Taught modules across all levels including courses on Databases, Multi Agent Systems, Logic, Statistics and Machine Learning.
- **Software Engineering Placement** - Royal Holloway University, Summer 2021
 - Worked with a fellow graduate to investigate the feasibility of transitioning our first year robotics course to an open source software platform.
- **Undergraduate Research Placement** - Royal Holloway University, Summer 2019
 - Worked in a small team investigating the use of Blockchain technology for supply chain management and carbon tracking.
- **Sales Assistant** - Maplin Electronics, 2016-2017
 - Customer facing role selling consumer electronics ranging from multimedia devices and domestic appliances to discrete components.

Education

- **PhD Student, Distributed and Intelligent Computing Environments Laboratory, 2022-23 (Withdrew)** - Royal Holloway University of London
 - Worked on formalising and implementing Multi-Agent simulations such as distributed task negotiation and evolutionary game theoretic models. Also studied single and multi agent learning tasks. Chose to withdraw due to a change in personal circumstances and a desire to work on more practical and impactful problems.
 - **MSc Artificial Intelligence (Distinction), 2021-22** - Royal Holloway University of London
 - Masters Thesis entitled "AI Planning using SAT/SMT and Bounded Model Checking", deriving a perspective of planning tasks as satisfiability problems which could then be encoded and solved by an off the shelf DPLL based solver like Z3. Awarded yearly Outstanding Project.
 - Distributed Systems, studied the fundamental principles of distruted computing networks including Elixir/OTP implementation of the consensus algorithms Paxos.
 - Data Analysis, intensive course covering wide array of Machine Learning techniques using R, Matlab and Python. Implemented ML algorithms such as AHC, Linear Regression and Random Forests from first principles.
 - **First Class Hons. BSc Computer Science, 2018-21** - Royal Holloway University of London
 - Worked as a team of 8 on a fullstack web development project following agile methodology, awarded best team.
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Projects

- **BigMagnet**, Python - 2024
 - Python tool for querying and aggregating code datasets. Handles pagination and concurrent execution of large request queries before downloading and extracting relevant files.
 - Originally intended to be used for generating a large Rust dataset for LLM tuning then extended to handle a range of search query parameters.
 - **WAMM (WhatsApp Markov Models)**, Python - 2022
 - Trains a word level n-gram Markov Model using raw WhatsApp chat history data. Handles both the cleaning and parsing of large history files and statistical analysis of extracted data.
 - **QBert**, Elm, Python - 2020
 - Experiments in Multi-Agent task-fulfilment with queueing situated in physical 2d space.
 - Python model which simulates the model and produces log files which can then be parsed by browser based visualizer tool written in Elm.
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Hobbies and Interests

- **Music**, I've been playing bass for just over a decade and listen to a wide range of music from jazz to shoegaze.
 - **Sport**, I play football, cycle and have recently started bouldering. When it comes to watching sport I mostly follow football but also try to keep up with rugby union, F1 and baseball.
 - **Gaming**, I love playing games with friends be it tabletop or virtual.
 - **Puzzles**, I enjoy puzzles in all their forms. I'm a devout Wordler and Rubik's Cube speedsolver.
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